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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: EET 305

Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

		Marks
1	Plot $x(2t)$, if $x(t) = u(-t+2) + u(t-3)$.	3
2	Find whether following signal is odd or even, $x(t) = t^4 + 4t^2 + 6$.	3
3	State and prove the time shifting property of Fourier Transform.	3
4	Derive the electrical analogous elements corresponding to elements of mechanical translational system with force voltage analogy.	3
5	Explain the concept of positive real functions.	3
6	Determine the impulse response of the system having transfer function, $H(s) = \frac{s}{(s-2)(s+3)}$	3
7	Define the process of sampling and state sampling theorem.	3
8	Write any three properties of z-transform.	3
9	Check the stability of the system represented by the following characteristic equation using Jury's test: $2z^4 + 7z^3 + 10z^2 + 4z + 1 = 0$	3
10	Obtain the transfer function of given discrete time system described by the difference equation: $y(n) - \frac{5}{4}y(n-1) + \frac{1}{6}y(n-2) = 2x(n)$	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11 a) Find the response of a system with an impulse response of $h(t) = u(t)$ for an input signal of $x(t) = e^{-\alpha t}\, u(t)$ — 8

b) Determine whether the following systems are time-invariant and linear. — 6

(i) $3\frac{d\,y(t)}{dt} + 5ty(t) = x(t)$

(ii) $\frac{d\,y(t)}{dt} + 10\sin y(t) = 2x(t)$

(iii) $y(t)\frac{d\,y(t)}{dt} + 10\,y(t) = 2x(t)$

12 a) For the signal, $x(t)$ shown in Fig.1, plot: (i) $x(t-1)$ (ii) $x(-3t+2)$ 8

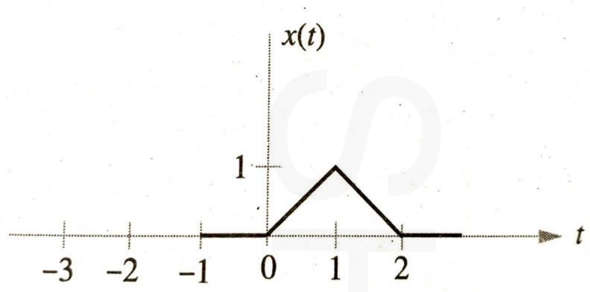

Fig.1

b) Test whether the following signals are periodic or not. If periodic, find the fundamental period. 6

(i) $x(t) = 5\sin 24\pi t + 7\sin 36\pi t$

(ii) $x(t) = e^{j(\pi t - 2)}$

(iii) $x(t) = 5\sin 24\pi t \cos 36\pi t$

Module -2

13 a) Determine the transfer function, $\frac{V_C(s)}{V_S(s)}$ of the given circuit. Hence, find the impulse response of the circuit. 4

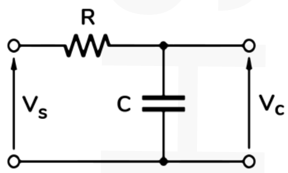

Fig.2

b) Obtain the trigonometric Fourier Series of the given signal 10

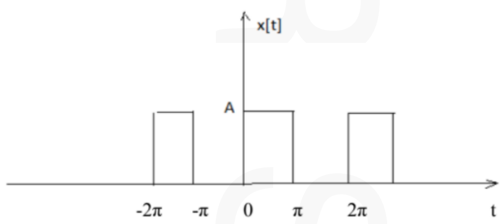

Fig.3

14 a) Derive the transfer function $\frac{X_2(s)}{F(s)}$ of the given mechanical system in Fig.4 10

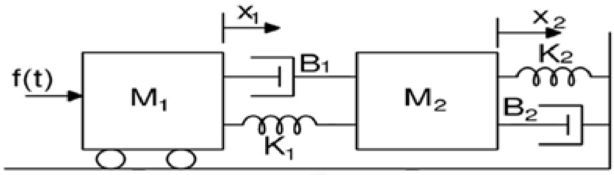

Fig.4

b) Find the transfer function of given LTI system. 4

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 2\frac{dx(t)}{dt} - 3x(t)$$

Module -3

15 a) How time response can be obtained from pole zero plot. 4

b) Describe Mason's gain formula for signal flow graph. Obtain the transfer function of the following system using Mason's gain formula. 10

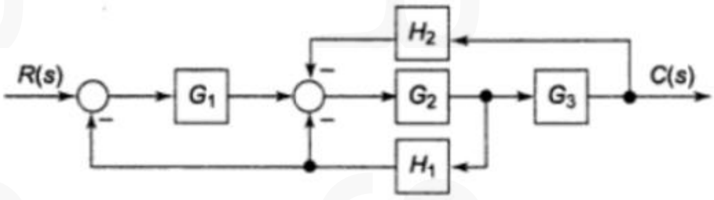

Fig.5

16 a) Check if the system with the given characteristic equation is stable using Routh Hurwitz stability criterion, $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$ 7

b) Obtain the transfer function for the following system using block diagram reduction technique. 7

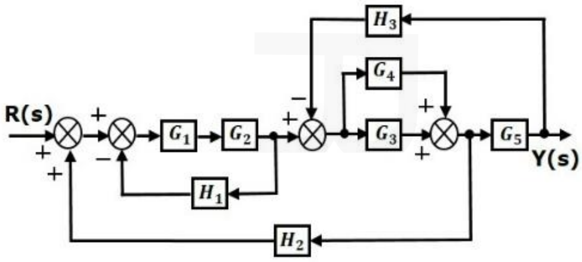

Fig.6

Module -4

17 a) Explain the significance of Zero order hold circuit in signal reconstruction. 4

b) Find the inverse Z-transform of: 10

$$X(Z) = \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}; ROC|Z| > 1$$

18 a) List any four properties of Discrete convolution. 4

b) Find the convolution sum of the following sequences 10

$$x(n) = 3\,\delta(n+1) - \delta(n) + 2\,\delta(n-1) + \delta(n-2)$$

$$h(n) = 2\,\delta(n) - 2\,\delta(n-1) + 2\,\delta(n-2) + \delta(n-3)$$

Module -5

19 a) State and prove the time reversal property of Discrete time Fourier transform. 4

b) A discrete system is given by the following difference equation: 10

$$y[n] - 5\,y[n-1] = x[n] + 4\,x[n-1]$$

where, $x[n]$ is the input and $y[n]$ is the output. Determine the expression for its magnitude response and phase response.

20 a) Check the stability of the given system using bilinear transformation: 6

$$F(z) = z^3 - 0.2z^2 - 0.25z + 0.05 = 0$$

b) Obtain the direct form-I and direct form-II realizations of the discrete time system with system function as: 8

$$H(z) = \frac{8 - 4Z^{-1} + 11Z^{-2} - 2Z^{-3}}{1 - \frac{5}{4}Z^{-1} + \frac{3}{4}Z^{-2} - \frac{1}{8}Z^{-3}}$$

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